



Data and Artificial Intelligence

Cyber Shujaa Program

Week 4 Assignment
Hotel Analytics Dashboard - Power BI Project

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1.0 Introduction

This project develops hands-on experience in Business Intelligence using Power BI for Hotel Management. Following the CodeBasics end-to-end data analyst project, this report documents the complete process of creating a hotel analytics dashboard and publishing it to the cloud.

The project utilizes the AtliQ Grands hotel chain dataset to create actionable insights for hospitality management decision-making. The project demonstrates proficiency in data loading, transformation, DAX calculations, visualization design, and cloud publishing.

Objectives

The objectives of this project are to:

- i. Understand the Hotel business and client needs
- ii. Load Data into Power BI
- iii. Transform Data using Power Query
- iv. Build DAX measures and calculations
- v. Visualize Dashboard for business insights
- vi. Publish project as part of portfolio collection

2.0 Task Completion

2.1 Understanding Hotel Business and Client Needs

Business Context: AtliQ Grands owns multiple five-star hotels across India. They have been in the hospitality business for the past 20 years. Due to strategic moves from other competitors and ineffective decision-making in management, AtliQ Grands are losing its market share and revenue in the luxury/business hotels category.

Client Requirements:

1. Create a dashboard to analyze revenue trends
2. Monitor occupancy rates across properties
3. Track ADR (Average Daily Rate) and RevPAR (Revenue per Available Room)
4. Analyze booking platforms performance
5. Identify weekly and monthly patterns
6. Compare performance across different cities

Key Metrics Requested:

- Revenue
- RevPAR (Revenue Per Available Room)
- ADR (Average Daily Rate)
- Occupancy %
- Realization %
- DSRN (Daily Sellable Room Nights)
- DBRN (Daily Booked Room Nights)
- DURN (Daily Utilized Room Nights)

2.2 LOAD DATA**Data Files Loaded:**

1. **dim_date.csv** - Contains date information
2. **dim_hotels.csv** - Property information for 7 hotels
3. **dim_rooms.csv** - Room categories (4 types: Standard, Elite, Premium, Presidential)
4. **fact_bookings.csv** - Booking transaction data
5. **fact_aggregated_bookings.csv** - Pre-aggregated daily metrics

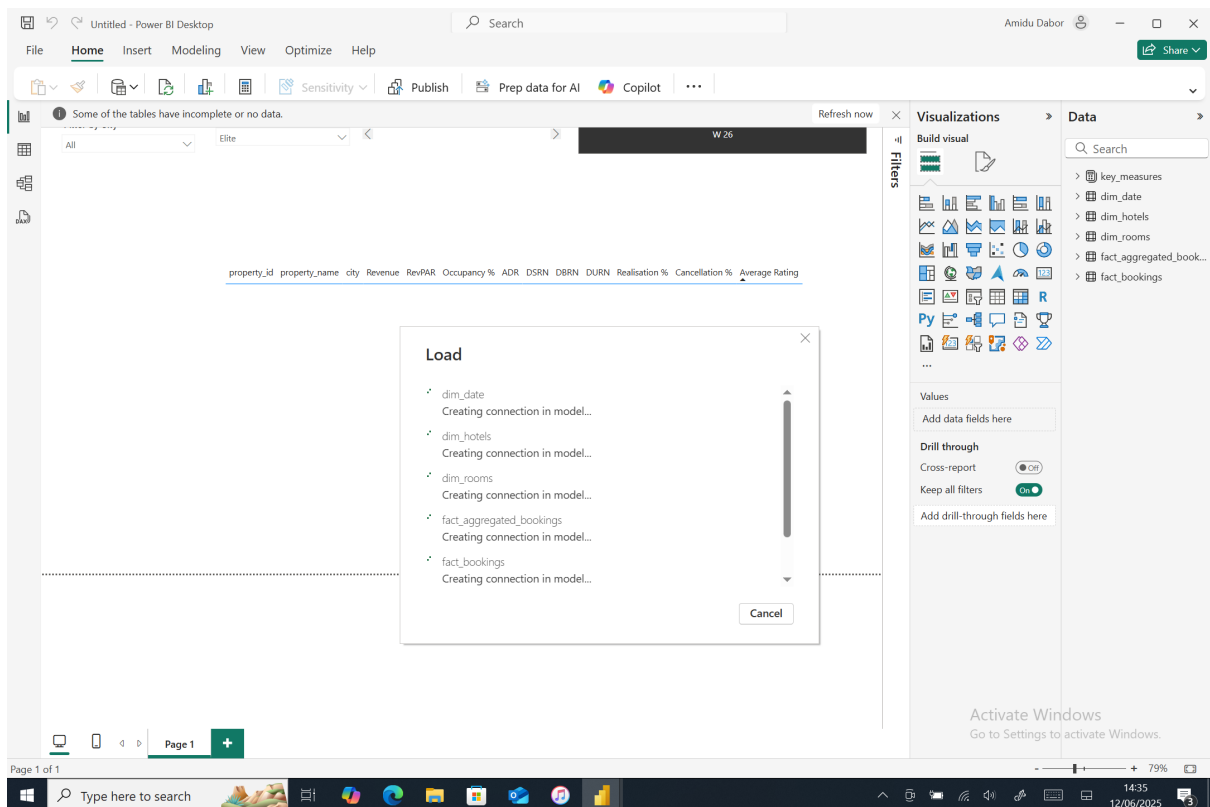
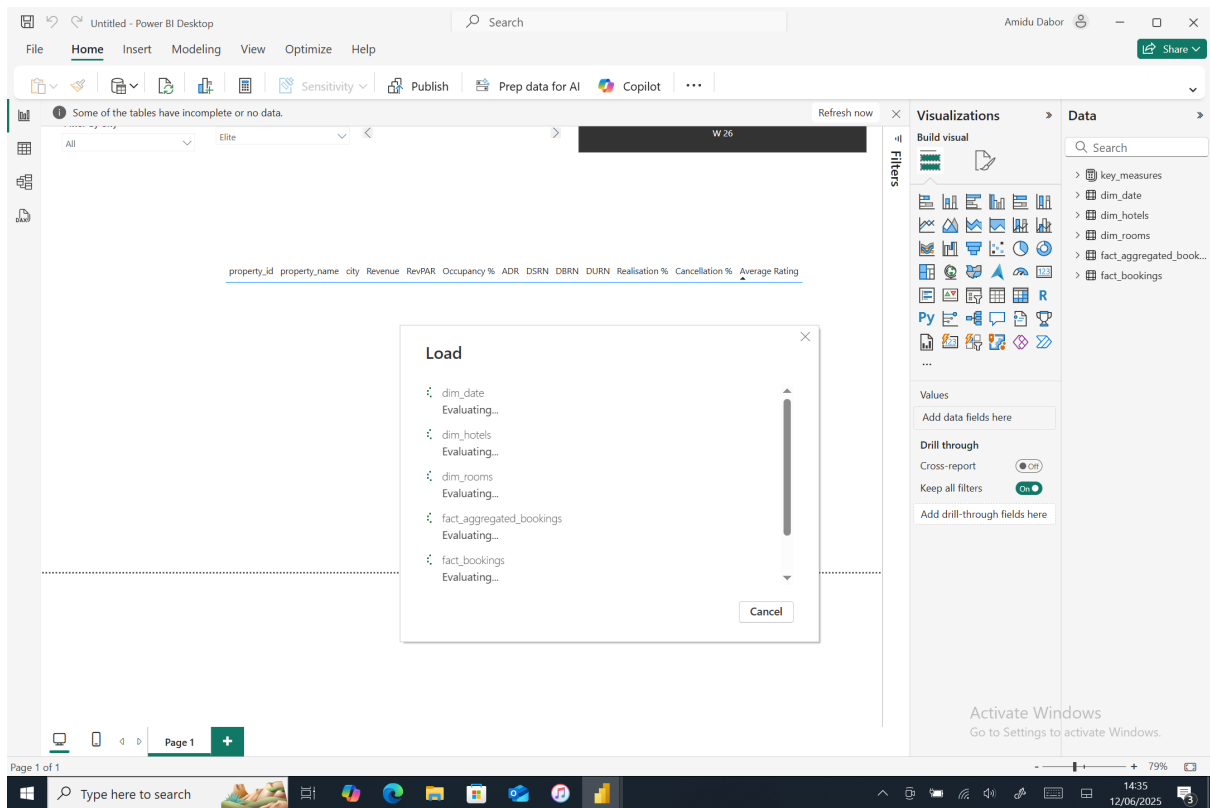
Loading Process:

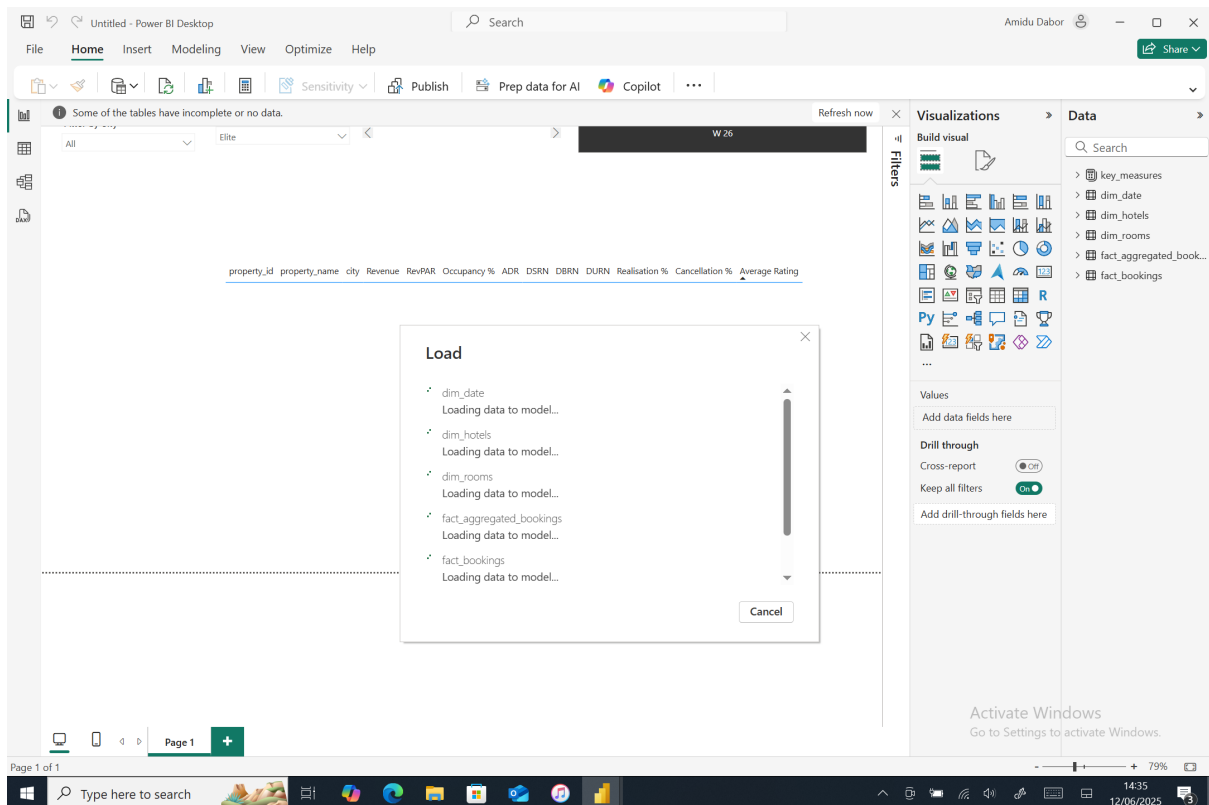
1. Opened Power BI Desktop
2. Selected "Get Data" → "Text/CSV"
3. Navigated to the data folder from CodeBasics
4. Selected all 5 CSV files using multi-select
5. Clicked "Load" to import into Power BI

Microsoft Power BI Desktop interface showing a file selection dialog box. The dialog title is "C:\Users\Amidu Daboh\Downloads\Power BI - Hospitality Management\Files to Use". The table below lists the files available for selection:

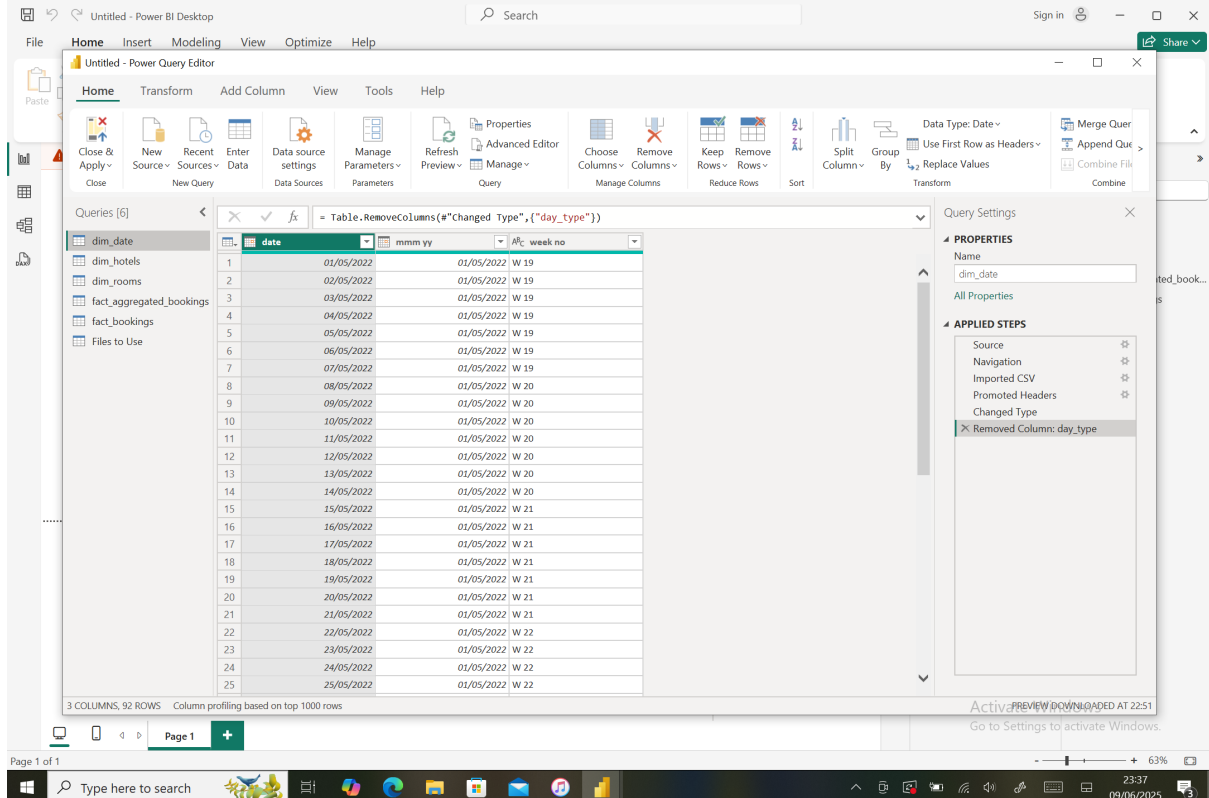
Content	Name	Extension	Date accessed	Date modified	Date created	Attributes	File path
Binary	dim_date.csv	.csv	09/06/2025 22:41:52	17/05/2023 12:26:42	09/06/2025 22:41:52	Record	C:\Users\Amidu Daboh\Downloads\Power BI - Hospitality Management\dim_date.csv
Binary	dim_hotels.csv	.csv	09/06/2025 22:41:52	17/05/2023 12:26:42	09/06/2025 22:41:52	Record	C:\Users\Amidu Daboh\Downloads\Power BI - Hospitality Management\dim_hotels.csv
Binary	dim_rooms.csv	.csv	09/06/2025 22:41:52	17/05/2023 12:26:42	09/06/2025 22:41:52	Record	C:\Users\Amidu Daboh\Downloads\Power BI - Hospitality Management\dim_rooms.csv
Binary	fact_aggregated_bookings.csv	.csv	09/06/2025 22:41:52	17/05/2023 12:26:42	09/06/2025 22:41:52	Record	C:\Users\Amidu Daboh\Downloads\Power BI - Hospitality Management\fact_aggregated_bookings.csv
Binary	fact_bookings.csv	.csv	09/06/2025 22:41:52	17/05/2023 12:26:42	09/06/2025 22:41:52	Record	C:\Users\Amidu Daboh\Downloads\Power BI - Hospitality Management\fact_bookings.csv
Binary	meta_data_hospitality.txt	.txt	09/06/2025 22:41:52	17/05/2023 12:26:42	09/06/2025 22:41:52	Record	C:\Users\Amidu Daboh\Downloads\Power BI - Hospitality Management\meta_data_hospitality.txt
Binary	metrics_list.xlsx	.xlsx	09/06/2025 22:41:52	17/05/2023 12:26:42	09/06/2025 22:41:52	Record	C:\Users\Amidu Daboh\Downloads\Power BI - Hospitality Management\metrics_list.xlsx

The dialog box includes buttons for "Combine", "Load", "Transform Data", and "Cancel". The background shows the Power BI Desktop ribbon with tabs like File, Home, Insert, Modeling, View, Optimize, and Help. The Data pane on the right lists the loaded tables: dim_date, dim_hotels, dim_rooms, fact_aggregated_bookings, and fact_bookings.





Loaded Tables



2.3 TRANSFORM DATA

Data Transformation Applied:

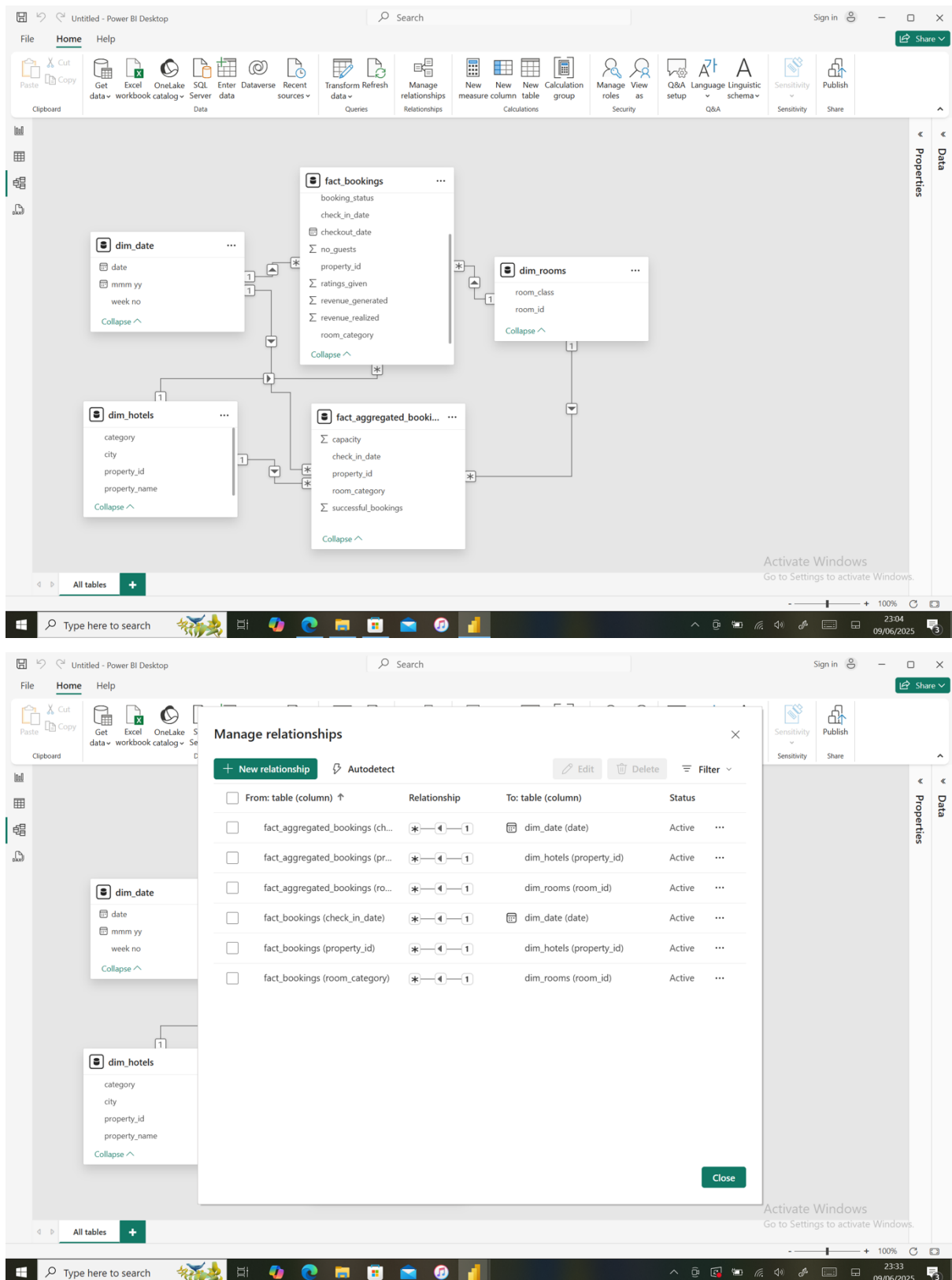
1. **Column Renaming**
 - Renamed raw column headers to more readable names (e.g., revpar → Revenue per Available Room, adr → Average Daily Rate).
2. **Data Type Conversion**
 - Converted columns like dates, integers, and decimals to appropriate data types.
3. **Removing Unnecessary Columns**
 - Removed irrelevant or unused columns to simplify the data model.
4. **Filtering Rows**
 - Applied filters to exclude data outside a specific range (e.g., Week 32 (W32)).
5. **Adding Calculated Columns**
 - Added new columns for metrics like Occupancy Rate, RevPAR, or custom groupings using **Add Column** → **Custom Column**.
6. **Formatting & Sorting**
 - Cleaned text fields (e.g., trimming whitespace, proper casing).
 - Sorted tables for clarity and reporting.
7. **Merging & Appending Queries**
 - Combined multiple data sources (e.g., room bookings + revenue) through **Merge Queries** or **Append Queries**.

2.4 BUILD DATA MODEL

Star Schema Implementation:

The data model follows a star schema pattern with two fact tables at the center and three dimension tables.

Relationships Created:



2.5 BUILDING METRICS DAX (DATA ANALYSIS EXPRESSION)

Key Measures Created:

1. Revenue Measures:

DAX

Revenue = SUM(fact_bookings[revenue_realized])

Total Bookings = COUNT(fact_bookings[booking_id])

2. Occupancy Measures:

DAX

Total Capacity = SUM(fact_aggregated_bookings[capacity])

Total Successful Bookings = SUM(fact_aggregated_bookings[successful_bookings])

Occupancy % = DIVIDE([Total Successful Bookings], [Total Capacity], 0)

3. ADR and RevPAR:

DAX

ADR = DIVIDE([Revenue], [Total Bookings], 0)

RevPAR = DIVIDE([Revenue], [Total Capacity], 0)

Alternative RevPAR calculation:

RevPAR Check = [ADR] * [Occupancy %]

4. Realization and Cancellation Metrics:

DAX

Realisation % = 1 - ([Cancellation %]+[No Show rate %])

Total Bookings = COUNT(fact_bookings[booking_id])

Cancellation % = DIVIDE([Total cancelled bookings],[Total Bookings])

5. Room Nights Metrics:

DAX

DSRN = DIVIDE([Total Capacity], [No of days])

DBRN = DIVIDE([Total Bookings], [No of days])

DURN = DIVIDE([Total Checked Out],[No of days])

6. Week-over-Week Changes:

DAX

Revenue WoW change % =

Var selv =

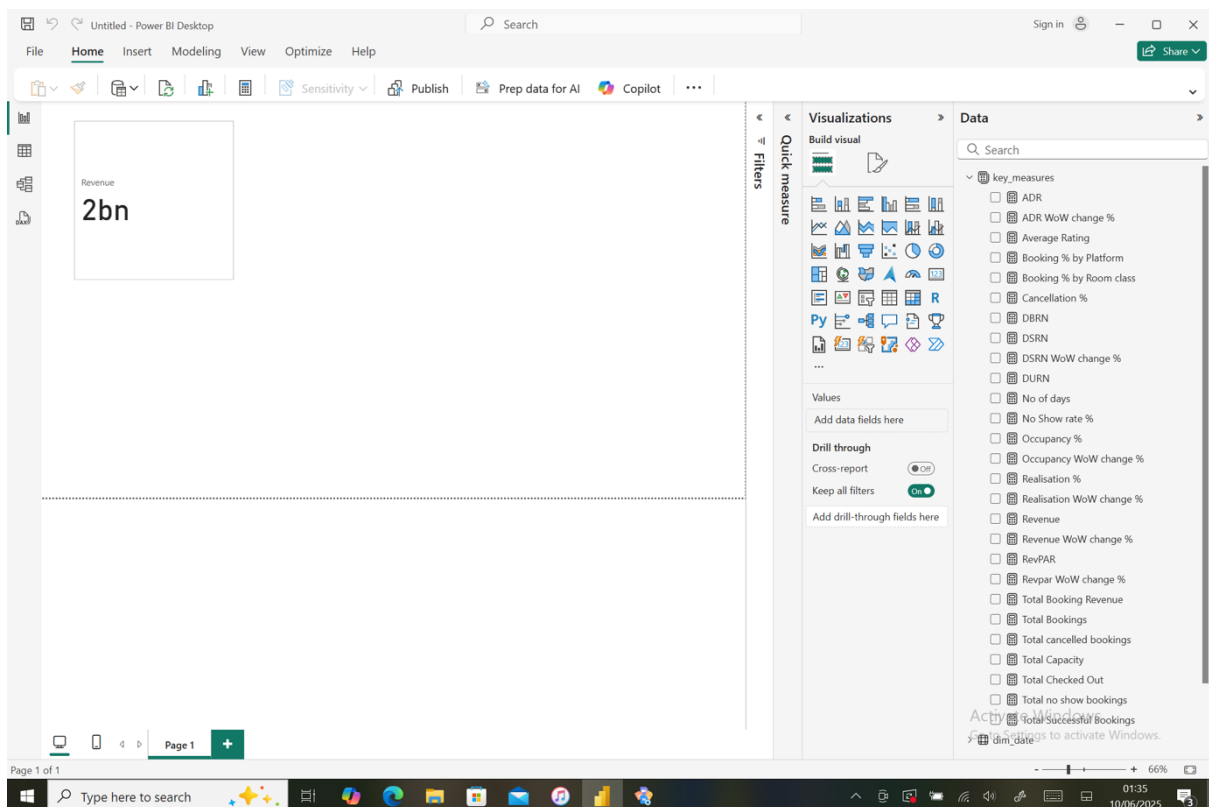
IF(HASONEFILTER(dim_date[wn]),SELECTEDVALUE(dim_date[wn]),MAX(dim_date[wn]))

var revcw = CALCULATE([Revenue],dim_date[wn]= selv)

var revpw = CALCULATE([Revenue],FILTER(ALL(dim_date),dim_date[wn]= selv-1))

return

DIVIDE(revcw,revpw,0)-1



2.6 VISUALIZE DASHBOARD

Dashboard Design:

Analytics Dashboard

1. Key Metrics Cards (Top Row):

- Revenue
- RevPAR

- ADR
- Occupancy %
- Realization %
- Cancellation %

2. Revenue by Category (Donut Chart)

- Luxury
- Business

3. Revenue Trend by Week (Line Chart)

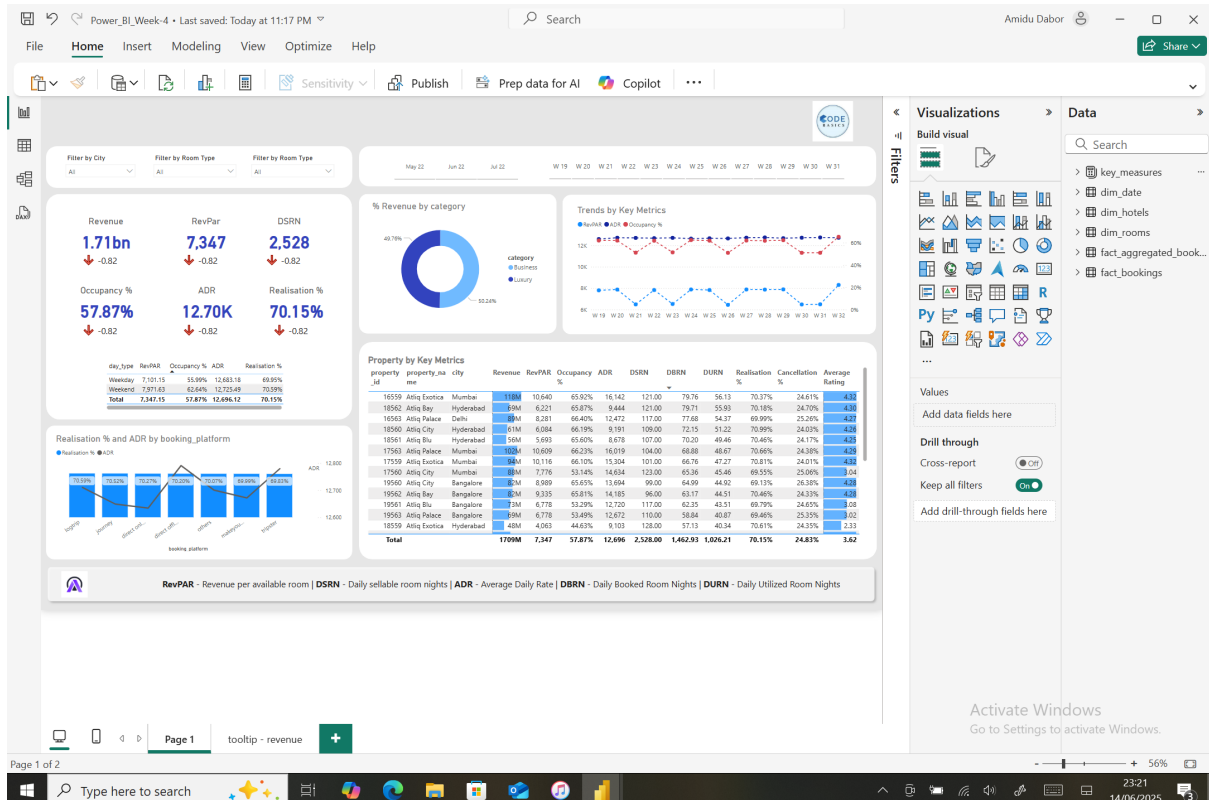
- X-axis: Week number
- Y-axis: Revenue
- Shows weekly revenue patterns

4. Property Performance Table

- Columns: Property Name, Revenue, Occupancy %, ADR, RevPAR, Realization %
- Sorted by Revenue descending

5. Filters Panel (Left Side):

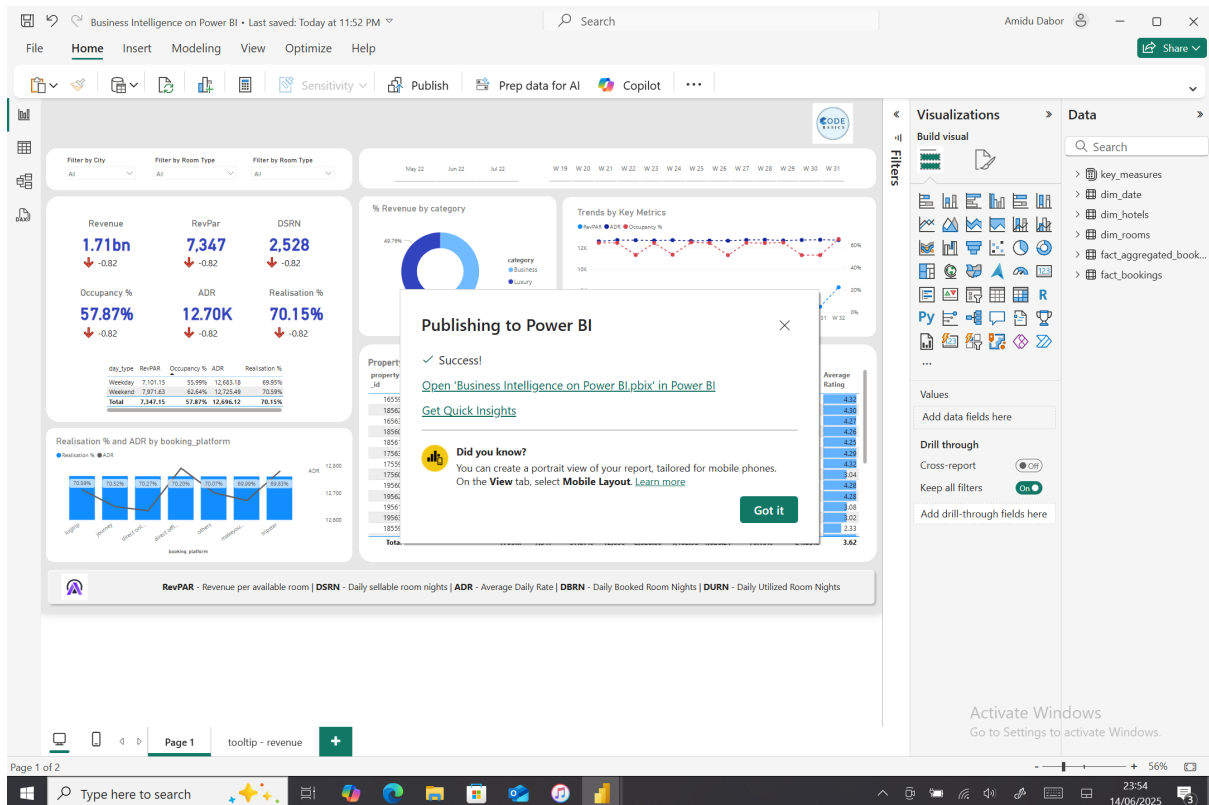
- City slicer
- Room class slicer
- Month selector
- Week selector

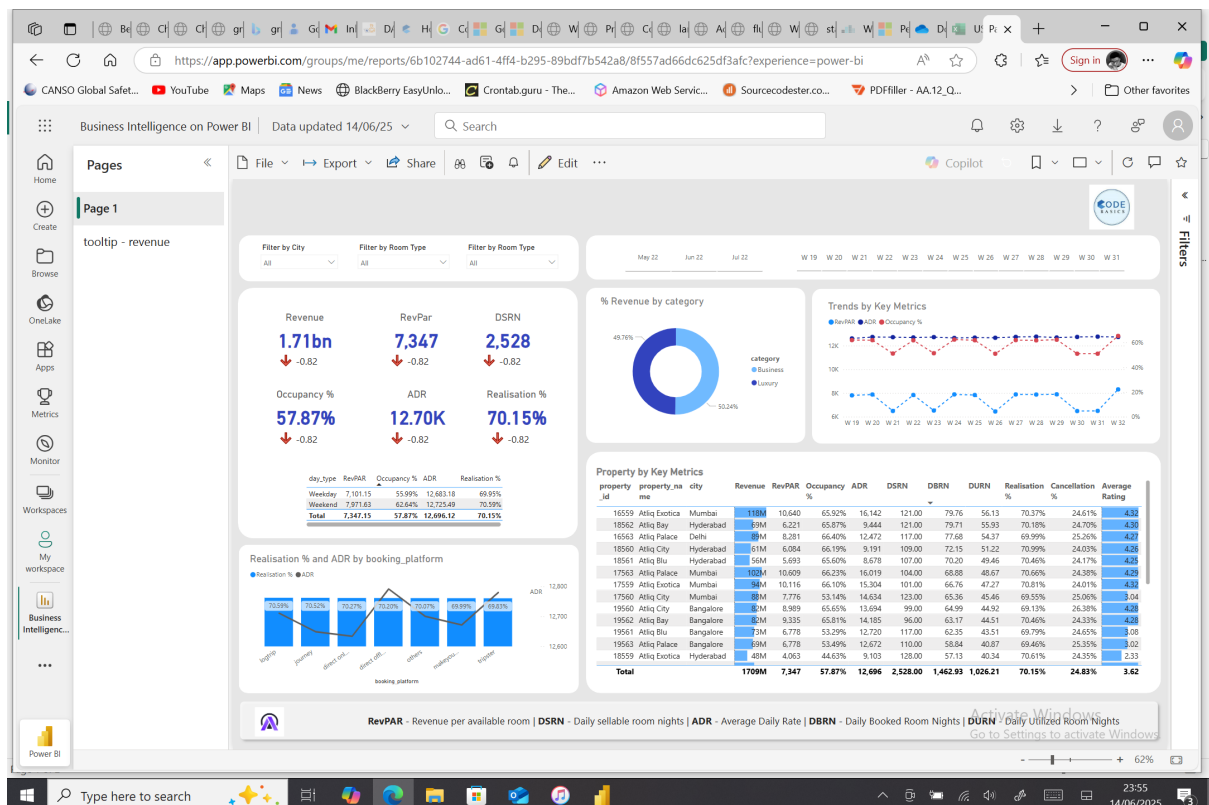


2.7 PUBLISH AND SHARE

1. Publish to Power BI Service:

- Clicked "Publish" in Power BI Desktop
- Selected "My workspace"
- Report name: "Business Intelligence on Power BI "
- Successfully published message received

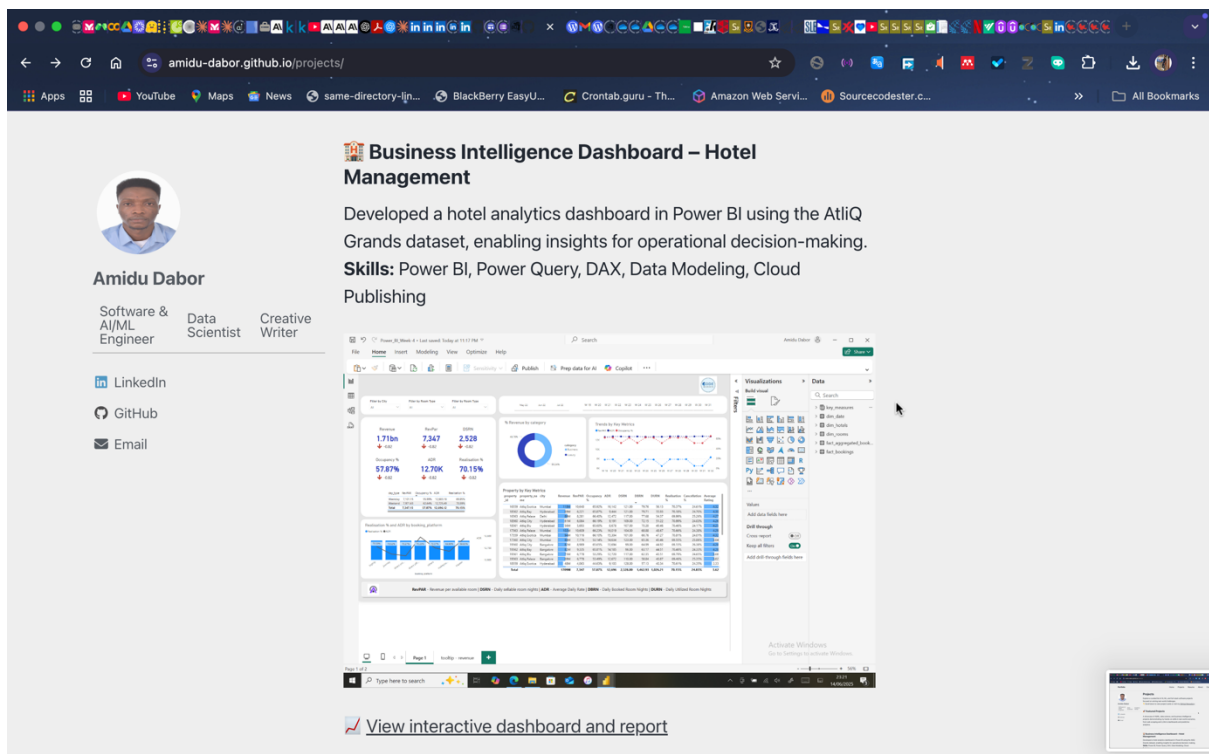
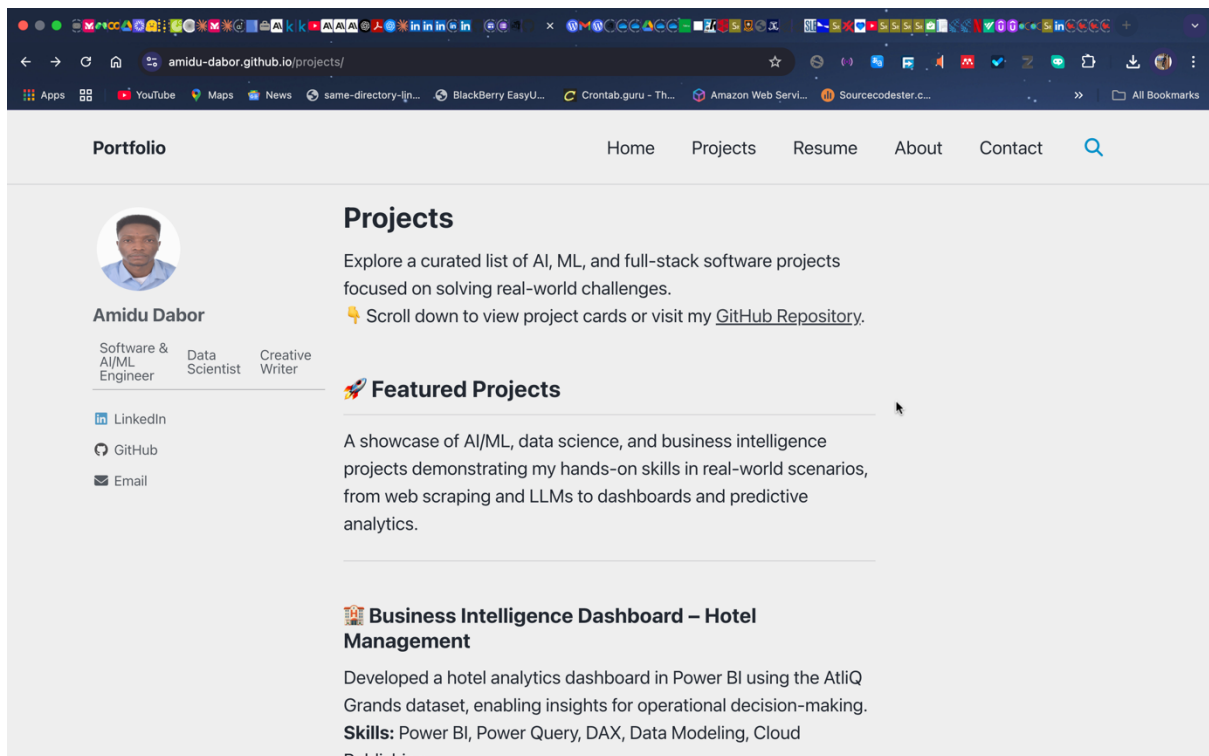




2. Public Access Link: Published Dashboard URL:

<https://drive.google.com/file/d/1HAOLujr0fVrfDQA8ucUDLyWcnF7MZ2P8/view?usp=sharing>

3.0 PORTFOLIO WEBSITE/HOTEL REVENUE ANALYTICS DASHBOARD - POWER BI PROJECT



4.0 CONCLUSION

This Business Intelligence project successfully achieved all assignment objectives:

1. **Understood Hotel Business Needs** - Analyzed AtliQ Grands' requirements for revenue and occupancy insights
2. **Loaded Data** - Successfully imported all 5 data files into Power BI
3. **Transformed Data** - Cleaned and prepared data using Power Query
4. **Built Data Model** - Implemented proper star schema with relationships
5. **Created DAX Measures** - Developed all required KPIs and calculations
6. **Visualized Dashboard** - Created interactive two-page dashboard
7. **Published and Shared** - Deployed to Power BI Service with public access

Technical Skills Demonstrated:

- Power BI Desktop proficiency
- Power Query M language
- DAX calculations
- Data modeling best practices
- Dashboard design principles
- Cloud publishing and sharing

Business Value Delivered: The dashboard enables AtliQ Grands management to:

- Monitor real-time revenue performance
- Identify underperforming properties
- Optimize pricing strategies
- Reduce cancellation rates
- Improve direct booking channels

Lessons Learned:

1. Importance of understanding business context before technical implementation
2. Value of proper data modeling for performance
3. Impact of effective visualizations on decision-making
4. Benefits of cloud publishing for accessibility

5.0 REFERENCES

1. CodeBasics. (2025). *End-to-End Data Analyst Project - Hotel Revenue Analytics*. Retrieved from <https://codebasics.io/resources/end-to-end-data-analyst-project>
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4. SQLBI. (2025). *DAX Patterns*. Retrieved from <https://www.daxpatterns.com/>